The Assignment 1 of Psychometric

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1. Introduction

In psychological research,measuring latent traits like impulsivity and psychological health is crucial for theoretical and practical purposes. Conventionally, Classical Test Theory (CTT) assesses test reliability based on overall scores, while Item Response Theory (IRT) provides a more detailed examination of individual test items. The study compares both approaches by using a dataset based on a large-scale survey of the social integration of young adults.

Firstly, we focused on Cronbach's alpha and omega, to examine the reliability of test items. Then, IRT modeling was applied to identify the best-fit model and estimate the difficulty and discrimination, assessing how well each item differentiates between individuals across trait levels. We also want compare the test information functions of logistic and linear models to observe the differences. Finally, we want to check for bias in test items across categories such as gender, study delay, and age.

2. Method

2.1. Dataset

The dataset collected in 1987 and 1991 is used in this analysis comes from a large-scale survey on the social integration of young adults, focusing on people aged 17 to 27. The sample includes 1775 participants who were measured by a series of psychological tests, considering as the variables. For the purposes of this analysis, we focused on the following variables:

- Impulsivity: Item scores on a test to measure the propensity to react fast and sometimes recklessly.
- General psychological health: Item scores on a test to measure the degree to which the subject suffers from minor psychiatric disorders.
- Gender: Whether the subject identifies as a man or a woman.
- Study delay: How many years the person is behind in his/her study.
- Age: Age in years.

To handle missing values, we can simply remove them and perform the analysis easily.

2.2. Statistical Analysis

2.2.1. Descriptive analysis

The descriptive statistics such as the mean, standard deviation, and range were computed for both scales. This initial analysis provides insights into the central tendency and spread of responses, giving a general sense of how the data is distributed among the participants.

2.2.2. Reliability Analysis

Cronbach's Alpha (α) and Omega (ω) were calculated for both impulsivity and psychological health to evaluate internal consistency. These measurements provide estimates of the reliability of each scale in measuring the corresponding psychological construct.

2.2.3. Factor Analysis

A one-factor model was used to measure factor loadings for the impulsivity and psychological health scales, assessing how well each item represented the latent trait.

2.2.4. IRT Model

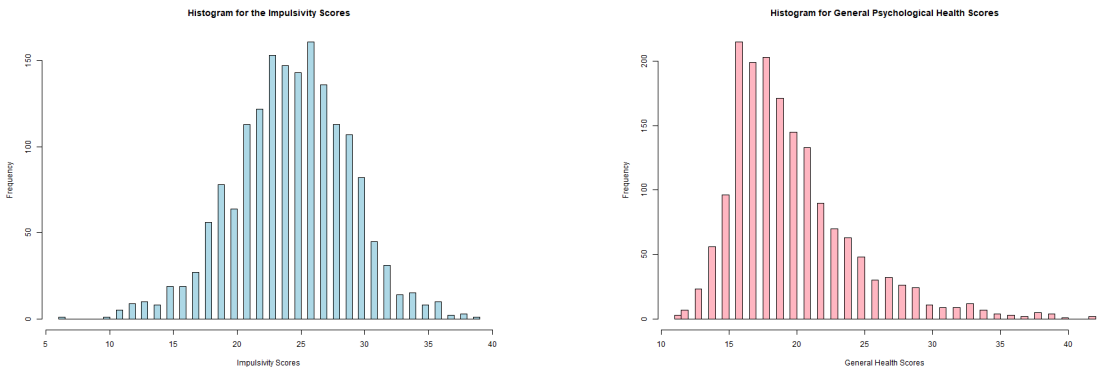
The IRT models was used to estimate item difficulty and discrimination on both scales in this analysis. We also plot the Test information of both scales to observe the precision of the test at various levels of the latent trait.

3. Results

3.1. Descriptive Statistics

Statistic	Impulsivity	Psychological_Health
Min	6.00	11.00
1st Qu.	22.00	17.00
Median	25.00	19.00
Mean	24.50	19.82
3rd Qu.	28.00	22.00
Max	39.00	42.00
Sd	4.52	4.58

Table 1. Descriptive Statistics for Impulsivity and Psychological Health



(a) The histogram of the Impulsivity Scores

(b) The histogram of the Psychological Health Scores

Figure 1. Distributions for Impulsivity and Psychological Health Scores

According to the table 1 and histograms(shown in Figure 1), impulsivity scores ranged from a minimum of 6 to a maximum of 39, with a median score of 25. The mean was slightly lower at 24.5 (table 1), suggesting a slight right skew in the distribution. This means that the majority of respondents are clustered at the lower end of the score range, while a smaller number of outliers at the upper end influence the mean, pulling the mean higher.The quartiles ranged from 22 to 28, indicating that most respondents scored within this range, with only a few showing extreme levels of impulsivity.

The left graph shows a unimodal distribution of impulsivity scores, with a clear peak around 25. The distribution appears approximately symmetrical and bell-shaped, consistent with the median calculated in the descriptive statistics. Most respondents fall within a range of scores from 17 to 30, which almost corresponds to the range between the 1st quartile (22) to the 3rd quartile (28). This indicates that the majority of scores are centrally concentrated, with fewer respondents at the extreme low and high ends of the scale, represented by clustering around the mean and median.

The right plot displays the distribution of general psychological health scores, which also appears unimodal with a peak around the score of 17. Unlike the impulsivity scores, the distribution is right-skewed, with most scores falling within the range of 13 to 25. Since lower scores indicate higher levels of mental health, this skewness suggests that there are more people with higher levels of mental health and fewer people with lower levels of mental health.This deviation represents the test selects more respondents with lower scores on the psychological health scale.

3.1.1. Reliabilty

Dataset	Cronbach-Guttman's Alpha	95% CI for Cronbach-Guttman's Alpha	McDonald's Omega
Impulsivity Scores	0.619	(0.588, 0.652)	0.313
Psychological Health Scores	0.810	(0.791, 0.831)	0.815

Table 2. Summary of Cronbach-Guttman's Alpha and McDonald's Omega

We use the bootstrapping method to resample the data 1000 times to generate a distribution of alpha values, which allows us to calculate the confidence interval(Table 2). The alpha for impulsivity test is around 0.619 (95% CI : 0.588–0.652), indicating a moderate reliability. The alpha for and general psychological health test is around 0.810 (95% CI: 0.791–0.831), representing a high reliability.

Then, we used the Standard Error of Measurement (SEM) to evaluate the precision of the impulsivity and general psychological health tests. The SEM for the impulsivity test was calculated to be 2.79, indicating a moderate measurement error associated with scores on this test. Alongside this, the 90% confidence interval estimate for it was approximately 19.92 to 29.08.This relatively wide interval suggests that the impulsivity test has low precision in measuring the trait because there is a large amount of error contributing to the variability in the scores.The psychological health test's lower SEM (1.99) indicates greater precision than the impulsivity test with a narrower 90% CI (16.55-23.08), suggesting less measurement error.

To determine how many parallel items need to be added or removed each of the tests to obtain a test score with reliability (Cronbach's alpha) of 0.80, we can use the

Test	Item	Uniqueness	Loading
Impulsivity	S255	0.799	0.448
	S264	0.772	0.477
	S269	0.917	-0.288
	S274	0.602	0.631
	S279	0.869	-0.361
	S292	0.697	0.550
General Psychological Health	S327	0.765	0.485
	S328	0.748	0.502
	S329	0.893	0.327
	S330	0.906	0.307
	S331	0.736	0.514
	S332	0.521	0.692
	S333	0.750	0.500
	S335	0.478	0.723
	S336	0.554	0.668
	S337	0.680	0.566
	S338	0.704	0.544

Table 3. The results of the one-factor models for both tests

Spearman-Brown formula. Applying the formula, we get a k for impulsivity test around 2.47, this means the length of the test needs to be multiplied by 2.47, i.e the new number of items is 15, there is about 9 parallel items which need to be added. New items should target areas where current items have high distinctiveness, such as S255, S264, and S279, to ensure that they better reflect the impulsivity construct.

For the general psychological health test, we get a k around 0.94, since it is smaller than 1, therefore, we remove 1 item from it. We remove item S330 had the lowest loading, which had a weak relationship with the common factor.

3.1.2. Factory analysis

we calculate the McDonald's Omega(ω) by modeling one-factor models where the factor has a variance of one. This provides a more accurate estimate of the tests' reliabilities and helps us better understand their internal consistency.

By evaluating the single-factor model of the six items of the impulsiveness test (Table 3), we found that items S255 (0.799), S264 (0.772), S292 (0.550), and S279 (0.869) had high uniqueness, indicating that a large part of the variance in these items could not be explained by the common factor impulsivity. Only S274 (0.602) had the lowest uniqueness, indicating that it contributed more to the factor.

Additionally, the highest loading was observed for S274 (0.631),indicating a strong relationship with the impulsivity, while S269 (-0.288) and S279 (-0.361) showed negative loadings, likely due to reverse coding issues.The single factor accounted for 22.4% of

the total variance (SS loadings = 1.342), which is relatively low. In terms of chis-square statistics, the value is 133.58 with a p-value 2.17e-24 , this indicates a poor fit for the one-factor model.

For the general psychological health test with 11 items, items S332 (0.521) and S335 (0.478) had the lowest uniqueness, indicating strong contributions. And items S329 (0.893) and S330 (0.906) showed higher uniqueness, indicating weak consistency with the latent construct. And the highest loadings were for S335 (0.723) and S332 (0.692), indicating these items had strong relationship with the psychological health factor. The lowest loading was observed for S330 (0.307). The model explained 29.7% of the variance in these items (SS loading = 3.266), this means the overall variance was slightly better captured than the impulsiveness test. Although the chi-square test with a significance level of $p < .001$ indicates a significant lack of fit, the most items had moderate or high loadings on the single factor, indicating that they fit reasonably well with the construct of mental health. According to the proportion of variance explained the outcome of this one-factor model is reasonable.

After computing the correlation between the test scores, and between the true scores on both tests, the observed correlation (i.e test scores correlation) is quite low, around at 0.0067, indicating a weak association between impulsivity and general psychological health in the sample. Additionally, the correlation between the true scores which is around 0.0095, indicating a negligible relationship between impulsivity and general psychological health after adjusting for measurement error.

3.1.3. IRT Analysis

As we compute the reliability assessments before, we can perform different advanced statistical modeling such as the Graded response model(GPM) and the Generalized partial credit model(GPCM), which both are item response theory models. The GRM is typically used for items with ordered response categories, while the GPCM generalizes this to allow varying response structures across items.

	GRM for Impulsivity Test	GPCM for Impulsivity Test
AIC	38053.23	38290.6
BIC	38283.45	38520.82
LogLik	-18984.61	-19103.3

Table 4. The outputs for GRM and GPCM on the impulsivity test.

	GRM for General Psychological Health Test	GPCM for General Psychological Health Test
AIC	32467.71	32382.12
BIC	32707.57	32621.98
LogLik	-16189.86	-16147.06

Table 5. The outputs for GRM and GPCM on the general psychological health test.

According to the results given, we observed that the GRM fits the data better than GPCM for the impulsivity test, because it has lower AIC and BIC values, alongside a less negative Log-likelihood compared to the GPCM.

In contrast, the GPCM fits the dataset better for the general psychological health test.

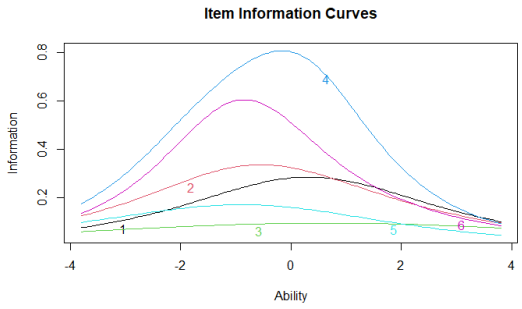


Figure 2. Item Information of GRM for impulsivity data

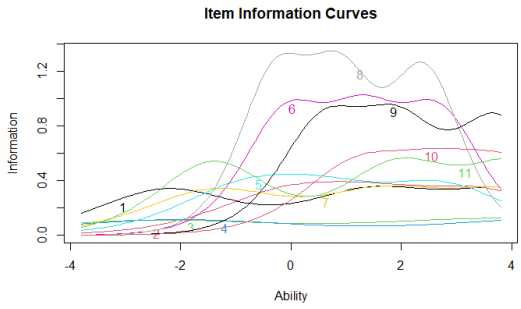


Figure 3. Item Information of GRM for general health data

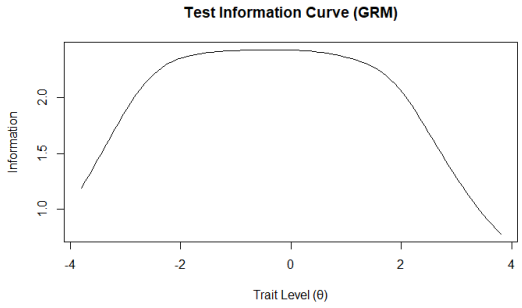


Figure 4. Test Information of GRM for impulsivity data

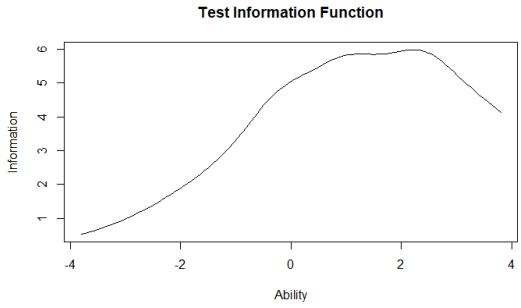


Figure 5. Test Information of GRM for general health data

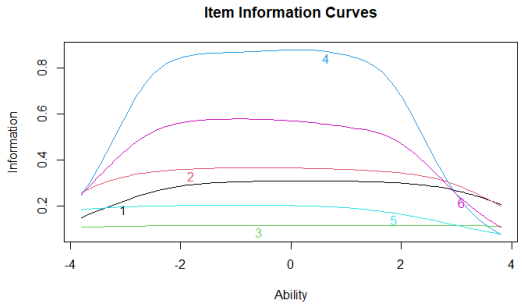


Figure 6. Item Information of GRM for impulsivity data

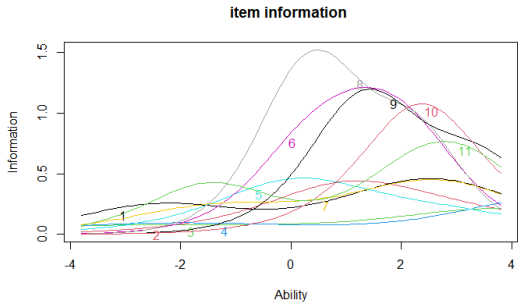


Figure 7. Item Information of GPCM for general health data

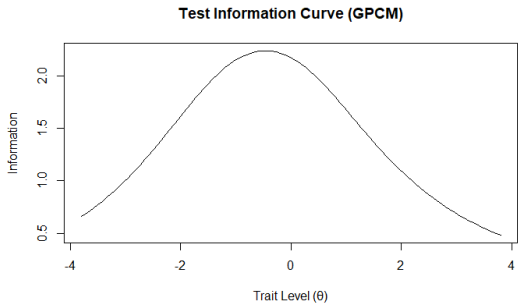


Figure 8. Test Information of GPCM for impulsivity data

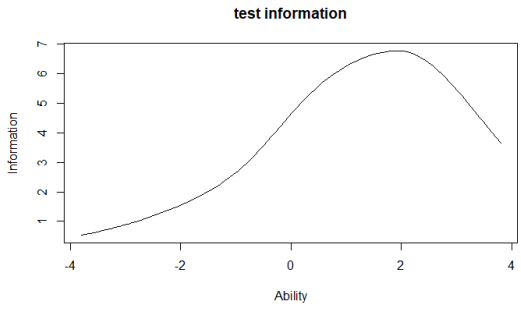


Figure 9. Test Information of GPCM for general health data

For the impulsivity test, the graded response model provided 6 categories (Extrmt1 to Extrmt6), item S269 has the highest value(Extrmt6 = 4.973), indicating that it requires a very high level of impulsivity for people to choose higher response categories. This makes S269 the most difficult item in the test. And the item S274 has the highest discrimination parameter with a value of 1.674, indicating it could effectively differentiate participants based on their impulsivity levels.

Additionally, the test information plots (Figures 4 and 8) of both the GRM and the GPCM for the impulsivity test showed that the test is most reliable for participants within the range round -2 to 2 on this scale. And the test is less effective at differentiating participants with very low or very high impulsivity(i.e below -2 or above +3).

For the general psychological health test, item S330 had the highest threshold value of 7.601 for category Extrmt3 in the GRM model and the highest value around 5.298 in the category 2 of the GPCM model, this indicates that it requires a very low level of psychological health for participants to select higher response categories. And item S335 had the highest discrimination value (Dscrmn = 2.217 in GRM and Dscrmn = 1.758 in GPCM), indicating that it has high validity in distinguishing participants with different levels of mental health.

Furthermore, the test information for both models was mostly between 0 and 3 on the ability scale, and the tests were most reliable in discriminating participants within this range. Compared to the GRM (Fig. 5), the GPCM plot (Fig. 9) obtains slightly higher peak information, making it slightly more effective in this range.

We also can perform IRT models with that of a model where the responses are completely independent. We could compare the chi-square:

$$\chi^2 = -2 \cdot (\text{log-likelihood of simpler model} - \text{log-likelihood of complex model})$$

The log-likelihood difference between the independence model and the GRM model for the impulsivity test is 1041.204, which is significant, indicating that the GRM model provides a better fit compared to the independence model. For the GPCM model, the log-likelihood difference is 893.954, which is significant. This means the GPCM model performs better than the null model. Overall, the GRM is the best-fitting model for the impulsivity test, and it provide the most accurate information.

For the general psychological test, the large chi-square statistics for both models suggest that both the GRM and GPCM provide a significantly better fit than the independence model. And the GPCM provides a slightly better fit than GRM for the general psychological test, because of its lower AIC around 32467.71, which is smaller than 3362.989, and its BIC value of 32621.98 that is smaller than 32707.57.

Now we would like to know more about whether the items of your primary test are biased with respect to a categorical background variable, such as gender. Therefore, we perform a Differential Item Function (DIF) analysis, and we use gender as our categorical variable in this case of impulsivity data.

We notice that the DIF analysis flagged 1 out of 6 items (item 5 corresponding to S279) in the impulsivity test as potentially biased with respect to gender. Item 5 showed potential DIF, especially for the thresholds corresponding to chi13 and chi23, suggesting

that participants from different groups may interpret to these thresholds differently. Additionally, items 1 to 4 and 6 did not show any evidence of DIF, as their chi-square values were relatively high and not significant.

This means that Item 5 may not measure impulsivity equally across genders, which could bias the results across genders.

In conclusion, we use different methods to assess the reliability, model-fitting for both tests. The GRM model generally provided a better fit for the impulsivity test, on the other hand, the GPCM model is better for the general psychological health test. We also perform the DIF to measure the influence of a categorical variable put in the dataset, the adjustments are needed to address item bias and improve measurement accuracy.

4. Discussion

The one-factor model, a latent variable model, assumes that the test measures a single underlying factor, allowing for more effective capture of measurement error compared to sum scores. This approach can provide a more accurate reflection of underlying traits. In contrast, IRT is more detailed, modeling the probability of specific item responses as a function of the latent trait. It offers precise estimates of individual ability or trait levels and accounts for item difficulty and discrimination, enhancing model fit and the understanding of how each item contributes to trait measurement. While sum scores are simple to compute and easy to interpret, they treat all items as equally important and fail to account for item difficulties or measurement error.

In research involving large groups, where the goal is to model the effects or associations of impulsivity or general psychological health with other variables, IRT is often preferred. This is because it can model item-level properties, offering a more accurate representation of constructs and enabling stronger associations with other variables.

In a clinical setting, where test results might guide decisions for an individual patient, sum scores are commonly chosen because of easier interpretation. However, if precision and accuracy are critical, a more sophisticated approach like the one-factor model may be a better option.